SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1717**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.*(BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 3

Duration : 1 Hour
Total Marks:30

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I Monika M / 192521435 *certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Monika M

Annexure A

Constraint-Based Problem Solving

1. A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that:

   Constraints:

   i. D1 cannot work Night shift

   ii. D2 must work before D3 (Morning < Afternoon < Night)

   iii. Only one doctor per shift

   iv. D3 cannot work Morning

   v. All doctors must be assigned exactly one shift

   Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule

2. A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G). Obstacles block certain paths.

   Constraints:

   i. Movement allowed: Up, Down, Left, Right
   ii. Cost of each move = 1
   iii. Cannot pass through obstacles
   iv. Use Manhattan Distance heuristic

   Using above constraints and BFS Search Algorithm Show step-by-step node expansion and priority queue updates and determine the optimal path and cost.

3. An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a partially observable environment where some paths are blocked, and it must make decisions with limited information.

The building is represented as a grid of rooms. The robot starts at position S and must reach a survivor located at G.

Constraints:

- The robot can move up, down, left, right
- Some cells are blocked (obstacles) and cannot be traversed
- The robot has limited sensing capability (can only observe adjacent cells)
- Each movement has a cost of 1, but entering risky zones has an additional cost of 2
- The robot must minimize total cost while ensuring safe navigation
- The robot must update its knowledge dynamically (online search scenario)

Tasks:

a) Analyze all the given constraints and explain how they affect the robot's decision-making process.

b) Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved.
c) Demonstrate how the robot applies AI concepts such as:

- Intelligent agents
- Rationality
- Environment type

d) Justify why your chosen approach is the most suitable for solving this problem under the given constraints.

ssssssRUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understandi n g	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritizatio n
Solution Developmen t	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justificatio n provided

1. Hospital Shift Scheduling (Backtracking Search)

This problem is a **Constraint Satisfaction Problem (CSP)**. We need to assign three doctors (D1, D2, D3) to three shifts (Morning, Afternoon, Night) while honoring specific rules.

The Constraints:

- **D1:** Cannot work Night.
- **D2 & D3:** D2 must work earlier than D3 (Morning < Afternoon < Night).
- **D3:** Cannot work Morning.
- **General:** One doctor per shift; every doctor gets one shift.
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Step-by-Step Backtracking:

1. **Assign D1:** Let's try the first available shift. **D1 = Morning**. (Valid, as D1 only refuses Night).

2. **Assign D2:** The Morning is taken. Let's try **D2 = Afternoon**.

3. **Assign D3:** Only Night is left. **D3 = Night**.

    4. **Constraint Check:**
        - Is D2 before D3? Yes (Afternoon < Night).
        - Is D3 NOT in the Morning? Yes (D3 is Night).
        - Is D1 NOT in the Night? Yes (D1 is Morning).

    5. **Final Valid Schedule**:
        - Morning: D1
        - Afternoon: D2
        - Night: D3.

2. Robot Grid Navigation (Search Algorithm)

The robot must move from Start (S) to Goal (G) in a 5x5 grid using a search strategy that incorporates the **Manhattan Distance heuristic**.

Key Parameters:

- **Heuristic:** Manhattan Distance = .
- **Priority Queue:** Used to expand the node with the lowest heuristic value first.
- **Cost:** Each move costs 1.

Expansion Steps (General Logic):

- **Step 1:** Start at node S. Calculate the Manhattan distance to G for all neighbors (Up, Down, Left, Right).

- **Step 2 (Priority Queue):** Add neighbors to the queue. The robot will always pick the neighbor that physically looks "closer" to the goal according to the grid coordinates.

- **Step 3:** If an obstacle is encountered, that path is discarded, and the robot picks the next best option from the queue.

- **Optimal Path & Cost:** The optimal path will be the one with the fewest moves (Cost = number of steps) that successfully maneuvers around obstacles to reach G.

3. **Autonomous Rescue Robot (Online Search Scenario)**

This scenario shifts from a known environment to a **partially observable** one where the robot must make decisions with limited information.

a) Constraint Analysis: The constraints (blocked paths, limited sensing, and "risky zones") mean the robot cannot plan a perfect path from the start. It must prioritize safety (minimizing cost) over speed. The **risky zone cost (+2)** forces the robot to favor longer, safer paths over shorter, dangerous ones.

b) Solution Approach: Online Search Agent Since the environment is only discovered as the robot moves, an **Online Search Agent** is most appropriate.

- **Steps:** 1. Sense immediate surroundings. 2. Update the internal map. 3. Use a strategy like **Uniform Cost Search (UCS)** to move to the lowest-cost adjacent cell. 4. If a dead-end is reached, backtrack and update the "knowledge base" to avoid that path again.

c) Application of AI Concepts:

- **Intelligent Agent:** The robot acts as a "percept-action" loop, sensing the building and moving toward survivors.
- **Rationality:** The robot is rational because it selects actions that maximize its performance measure (finding the survivor) while minimizing costs (avoiding risky/blocked zones).
- **Environment Type:** The environment is **partially observable** (cannot see through walls), **dynamic** (survivor locations or hazards might change), and **sequential** (current moves affect future options).

d) Justification: An **Online Search** approach is the most suitable because traditional "offline" algorithms (like BFS or A*) assume the entire map is known beforehand. In a disaster zone, information is gathered in real-time; the robot must be able to "learn" and adapt its path dynamically as it discovers obstacles.